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PROPYL TABLET

BOXED WARNING

Severe liver injury and acute liver failure, in some cases fatal, have been reported in patients treated with propylthiouracil. These reports of hepatic reactions include cases requiring liver transplantation in adult and paediatric patients.

Propylthiouracil should be reserved for patients who cannot tolerate carbimazole/methimazole and in whom radioactive iodine therapy or surgery are not appropriate treatments for management of hyperthyroidism.

Because of the risk of foetal abnormalities associated with carbimazole/methimazole, propylthiouracil may be the treatment of choice when an antithyroid drug is indicated during or just prior to the first trimester of pregnancy (See Warnings and Precautions).

DESCRIPTION

White round biconvex tablet with a bisect and embossed with "PROPYL", "SPS" on one side, plain on the other.
Each **Propyl** tablet contains 50mg Propylthiouracil USP.

PHARMACODYNAMICS

Within the thyroid gland, **Propyl** depresses the formation of iodotyrosines by inhibiting the iodination of thyroglobulin, and reduction of the synthesis of triiodothyronine and thyroxine by inhibiting the coupling reaction between iodotyrosines. In addition, **Propyl** blocks the peripheral deiodination of thyroxine (T_4) to the more potent triiodothyronine (T_3). **Propy**l does not interfere with the action and release of thyroid hormones already formed. Therefore, patients may only respond to **Propy**l after depletion of circulating and colloid-stored thyroid hormones. The rapid fall in serum triiodothyronine (T_3) concentration, before serum thyroxine (T_4) levels fall, parallels a clinical improvement in the thyrotoxic patient, and is generally seen after the first week. The patient may become euthyroid after 4-6 weeks. Hence, a decreased rate of conversion of circulating thyroxine (T_4) to triiodothyronine (T_3) may be clinically beneficial in patients with thyrotoxic crisis, since metabolic activity of triiodothyronine (T_3) is 3 to 5 times that of thyroxine (T_4).

PHARMACOKINETICS

Absorption: Propylthiouracil is rapidly absorbed from the gastrointestinal tract and has a bioavailability of 50-75%.
Distribution: Propylthiouracil appears to be concentrated in the thyroid gland. It readily crosses the placenta and is distributed into breast milk. About 80% of propylthiouracil is protein bound.
Metabolism: Propylthiouracil undergoes rapid first-pass metabolism in the liver where it is metabolised to its glucuronic acid conjugate.
Excretion: The elimination half-life of propylthiouracil is estimated to be 1-2 hours. Propylthiouracil is mainly excreted in the urine as the glucuronic acid conjugate. Very little unchanged drug is excreted in the urine and negligible amounts are excreted in the faeces.

INDICATION

Propyl is indicated for the treatment of hyperthyroidism or in the treatment of the thyrotoxic patient prior to surgery or radioactive-iodine therapy.

DOSAGE

Adults

Depending on the severity of the disorder, the usual initial dose is 300 to 600mg daily in divided doses, at 8-hourly intervals. Patients with severe hyperthyroidism and/or very large goitre may require initial doses of 600 to 1,200mg daily in divided doses.
Maintenance dose: 50 to 200mg daily when the condition is controlled, probably within 1 to 3 months.
Thyrotoxic crisis: Concomitant with the administration of other agents, e.g. iodides, adrenergic blocking agents, and general supportive measures, the recommended dose of **Propyl** is 200mg every 4 to 6 hours on the first day to be administered orally or by nasogastric tube. Once full control of symptoms is achieved, dosage is gradually reduced to the usual maintenance level. lodides (e.g. potassium iodide, strong iodine solution) are given to inhibit the release of hormone from the gland but may subsequently be used as a substrate for thyroid hormone synthesis. Therefore, treatment with **Propyl** is usually initiated before iodide therapy.

Children

6 to 10 years – 50 to 150mg daily in divided doses;
Over 10 years – 150 to 300mg or 150mg/m² daily.
Maintenance dose: 50 to 100mg daily, according to patient's response.
For the treatment of hyperthyroidism in neonates: 5 to 10mg/kg daily in divided doses has been recommended.

CONTRAINDICATIONS

Patients who are known to be hypersensitive to propylthiouracil or related thioamide derivatives.

WARNINGS AND PRECAUTIONS

Patients should be closely supervised during prolonged propylthiouracil therapy because of the likelihood of agranulocytosis. Patients should be warned to report immediately any evidence of illness, particularly sore throat, skin eruptions, fever, chills, headache and malaise. Therapy with **Propyl** should be discontinued and appropriate supportive and symptomatic therapy initiated if agranulocytosis, aplastic anaemia, hepatitis, fever or exfoliative dermatitis occurs. All patients receiving **Propyl** should have regular full blood counts as well as close monitoring of liver and thyroid function tests.

Hepatotoxicity resulting in liver failure, liver transplantation, or death, has been reported with propylthiouracil therapy in adults and children. Discontinue propylthiouracil if clinically important evidence of abnormal liver function occurs. Patients should be informed of the potential risk of serious hepatoxicity or liver injury including liver failure and death. Patients who are initiated with propylthiouracil should be closely monitored for signs and symptoms of liver injury (e.g. fatigue, weakness, vague abdominal pain, loss of appetite, itching, easy bruising or yellowing of the eyes or skin) especially during the first 6 months. If liver injury is suspected, promptly discontinue propylthiouracil therapy.

Propylthiouracil should not be used in paediatric patients unless the patient is allergic to or intolerant of the alternatives available.

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Another serious side effect is systemic vasculitis which can occur anytime and up to several years after initiation of treatment with propylthiouracil. Risk of systemic vasculitis may increase with prolonged use. Renal involvement is most common but skin, lung and musculoskeletal systems may also be involved. In severe cases death can occur. Propylthiouracil should be discontinued promptly and treatment initiated as required.

Prolonged therapy and/or excessive doses of propylthiouracil may cause hypothyroidism so thyroid function should be monitored regularly. Regular thyroid function tests are recommended in patient monitoring (recommended prior to initiation of therapy, at monthly intervals during stabilization, then every 2 to 3 months) viz. free (unbound) serum thyroxine (T_4) levels, total serum T_4 levels, serum thyrotropin (TSH), total serum triiodothyronine (T_3). Liver function tests are also recommended at periodic intervals during therapy.

Although propylthiouracil is used for the total treatment of hyperthyroidism, duration of treatment necessary to produce a prolonged remission varies from 6 months to several years, with an average duration of one year. Remission has occurred in at least 50% of patients 6-12 months after cessation of medication.

In preparing patients for surgery, the administration of iodine is recommended concomitantly with propylthiouracil to decrease the vascularity and friability of the thyroid gland. Propylthiouracil may cause hypothrombinaemia and bleeding so prothrombin time should be monitored during therapy, especially prior to surgery.

In view of the fact that hypothyroid patients seem to have poor adrenergic nervous function, use with caution in patients with asthma.

Effects on ability to drive and use machines
Propylthiouracil has no documented effects on the ability to drive or use machines.

PREGNANCY AND LACTATION

Women of childbearing potential
Women of childbearing potential should be informed about the potential risks of propylthiouracil use during pregnancy.

Use in pregnancy
Hyperthyroidism in pregnant women should be adequately treated to prevent serious maternal and foetal complications. Propylthiouracil is able to cross the human placenta. Animal studies are insufficient with respect to reproductive toxicity. Epidemiological studies provide conflicting results regarding the risk of congenital malformations. Individual benefit/risk assessment is necessary before treatment with propylthiouracil during pregnancy. Propylthiouracil should be administered during pregnancy at the lowest effective dose without additional administration of thyroid hormones. If propylthiouracil is used during pregnancy, close maternal, foetal and neonatal monitoring is recommended.

Use in lactation
Propylthiouracil is excreted in breast milk, as such, **Propyl** is contraindicated during nursing.

DRUG INTERACTIONS

Because propylthiouracil can cause hypoprothrombinaemia, extreme caution is advised in patients receiving oral anticoagulants or heparin. Prothrombin times should be carefully monitored during therapy. Concurrent use of propylthiouracil with agranulocytosis producing medications may potentiate the risk of agranulocytosis. The response of the thyroid gland to propylthiouracil may be impaired by a concurrent high iodine intake. Drug induced changes in thyroid status may affect the dosage requirements for theophylline and digitalis. The doses of digitalis and theophylline may need to be reduced as thyroid function returns to normal.

ADVERSE REACTIONS

Incidence of adverse reactions is dose-related and occurs most commonly during the first 2 months of therapy. The overall incidence of side effects with **Propyl** is of the order of 3%. Among the minor and most frequently encountered side effects are skin rashes including maculopapular eruptions, urticaria, pruritus, nausea, vomiting, gastric distress and headache. The following side effects have also been reported: drowsiness, neuritis, vertigo, loss of taste, arthralgia, myalgia paraesthesia, alopecia, skin pigmentation, oedema, lupus-like syndrome, vasculitis, hepatitis and nephritis.
Agranulocytosis usually occurs during the first 2 months of therapy and then the incidence gradually declines. Patients should be instructed to contact their doctor and report immediately any signs or symptoms of illness, particularly sore throat, skin eruption, fever, chills, headache and malaise. Thrombocytopenia, leucopoenia and aplastic anaemia have also been reported.

OVERDOSAGE AND TREATMENT

Symptoms of propylthiouracil overdose include: nausea, vomiting, epigastric distress, headache, fever, arthralgia, pruritus, oedema and pancytopenia, exfoliative dermatitis and hepatitis have occurred. Agranulocytosis is the most severe potential adverse effect due to acute propylthiouracil toxicity. Hypothyroidism may result from prolonged therapy.

Treatment of overdosage may consist of gastric lavage, observation, symptomatic and supportive therapy. If bone marrow depression develops, treatment by way of transfusion of fresh whole blood, antibiotics, and corticosteroids are used. Prothrombin deficiency associated with a haemorrhagic diathesis may be controlled by the administration of phytomenadione.

PACK SIZE

Box of 50 x 10 tablets.

STORAGE

Store in a cool dry place, below 30°C.
Protect from light.
Keep out of the reach of children.
Jauhi daripada kanak-kanak.

Shelf life: Please refer to the outer box label.

Manufactured by:

SPS SRIPRASIT
SRIPRASIT PHARMA CO., LTD.
216 Moo 6, Suanluang, Krathum Baen,
Samut Sakhon 74110, Thailand.

Date of revision: Aug 2022

PIP0005 Rev:5

หลัง

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